THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET 1

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 6	Chord $= 2\sqrt{r^2 - d^2}$.
2	**(a)** 5	$r^2 = h^2 + d^2$.
3	**(c)** 58°	Isosceles triangle OAB.
4	**(c)** 72°	$\angle A + \angle C = 180^\circ$.
5	**(b)** 48°	Central angle is double.
6	**(a)** 135°	Opposite angles of the cyclic quadrilateral are supplementary.
7	**(a)** 12	Equal chords are equidistant from the centre.
8	**(c)** 66°	Right angle at C.
9	**(c)** 12	Chord $= 2\sqrt{r^2 - d^2}$.
10	**(a)** 10	$r^2 = h^2 + d^2$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Half-chord $= \sqrt{20^2 - 16^2} = 12$ cm (1); Chord $= 24$ cm (1).	1+1
12	$\angle C = 94^\circ$ (1); $\angle D = 81^\circ$ (1).	1+1
13	$\angle OAB = \frac{180-68}{2} = 56^\circ$ (1); $\angle ACB = 34^\circ$ (1).	1+1
14	Half-chord $= \sqrt{25^2 - 15^2} = 20$ cm (1); Chord $= 40$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Distances from the centre: 15 cm and 8 cm (1); Same side: subtract (1); Distance $= 7$ cm (1).	1+1+1
16	$\angle A + \angle C = 180^\circ$; $5x = 170$ (1); $x = 34$ (1); $\angle A = 122^\circ$, $\angle C = 58^\circ$ (1).	1+1+1
17	$\angle OAB = \frac{180^\circ - 72^\circ}{2} = 54^\circ$ (1); $\angle ACB = \frac{1}{2}\angle AOB = 36^\circ$ (1); $\angle ADB = 180^\circ - 36^\circ = 144^\circ$ (1).	1+1+1
	OR: Distances from the centre: 24 cm and 7 cm (1); Same side: subtract (1); Distance $= 17$ cm (1).	1+1+1
18	$\angle A + \angle C = 180^\circ$; $6x = 174$ (1); $x = 29$ (1); $\angle A = 128^\circ$, $\angle C = 52^\circ$ (1).	1+1+1
	OR: $\angle OAB = \frac{180^\circ - 88^\circ}{2} = 46^\circ$ (1); $\angle ACB = \frac{1}{2}\angle AOB = 44^\circ$ (1); $\angle ADB = 180^\circ - 44^\circ = 136^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Half-chord $= 12$ cm (1); Distance $d = \sqrt{13^2 - 12^2} = 5$ cm (1); Area $= \frac{1}{2} \times 24 \times 5$ (1); $= 60$ cm^2 (1); Base is the chord, height is the perpendicular (1).	1+1+1+1+1
	OR: $d_1 = \sqrt{13^2 - 5^2} = 12$ cm (1); $d_2 = \sqrt{13^2 - 12^2} = 5$ cm (1); The chord of length 24 cm is nearer to the centre (1); A longer chord lies nearer to the centre (2).	1+1+1+2
20	Half-chord $= 12$ cm (1); Distance $d = \sqrt{15^2 - 12^2} = 9$ cm (1); Area $= \frac{1}{2} \times 24 \times 9$ (1); $= 108$ cm^2 (1); Base is the chord, height is the perpendicular (1).	1+1+1+1+1
	OR: $d_1 = \sqrt{17^2 - 8^2} = 15$ cm (1); $d_2 = \sqrt{17^2 - 15^2} = 8$ cm (1); The chord of length 30 cm is nearer to the centre (1); A longer chord lies nearer to the centre (2).	1+1+1+2

Mistakes to avoid:

- Using the full chord instead of the half-chord in $r^2 = d^2 + (\text{half-chord})^2$.
- Forgetting that the perpendicular distance, not the slant distance, is used when finding the distance of a chord from the centre.
- Applying the angle theorems to an arc without checking the angle is at the *centre* (double) or on the *remaining* circle (half).
- For two parallel chords, adding the distances when they are on the same side or subtracting them when they are on opposite sides (the reverse is correct).
- Assuming that opposite angles of a cyclic quadrilateral are equal; they are supplementary (sum 180°).
- Using the angle in a semicircle theorem when the chord is not a diameter.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 48°	Isosceles triangle OAB.
2	**(c)** 80°	$\angle A + \angle C = 180^\circ$.
3	**(c)** 56°	Central angle is double.
4	**(b)** 125°	Opposite angles of the cyclic quadrilateral are supplementary.
5	**(d)** 12	Equal chords are equidistant from the centre.
6	**(c)** 52°	Right angle at C.
7	**(c)** 10	Chord $= 2\sqrt{r^2 - d^2}$.
8	**(d)** 13	$r^2 = h^2 + d^2$.
9	**(d)** 38°	Isosceles triangle OAB.
10	**(a)** 60°	$\angle A + \angle C = 180^\circ$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle C = 107^\circ$ (1); $\angle D = 66^\circ$ (1).	1+1
12	$\angle OAB = \frac{180-92}{2} = 44^\circ$ (1); $\angle ACB = 46^\circ$ (1).	1+1
13	Half-chord $= \sqrt{29^2 - 20^2} = 21$ cm (1); Chord $= 42$ cm (1).	1+1
14	$\angle C = 58^\circ$ (1); $\angle D = 113^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Distances from the centre: 20 cm and 15 cm (1); Same side: subtract (1); Distance $= 5$ cm (1).	1+1+1
16	$\angle A + \angle C = 180^\circ$; $8x = 176$ (1); $x = 22$ (1); $\angle A = 120^\circ$, $\angle C = 60^\circ$ (1).	1+1+1
17	$\angle OAB = \frac{180^\circ - 100^\circ}{2} = 40^\circ$ (1); $\angle ACB = \frac{1}{2}\angle AOB = 50^\circ$ (1); $\angle ADB = 180^\circ - 50^\circ = 130^\circ$ (1).	1+1+1
	OR: Distances from the centre: 8 cm and 6 cm (1); Same side: subtract (1); Distance $= 2$ cm (1).	1+1+1
18	$\angle A + \angle C = 180^\circ$; $6x = 174$ (1); $x = 29$ (1); $\angle A = 72^\circ$, $\angle C = 108^\circ$ (1).	1+1+1
	OR: $\angle OAB = \frac{180^\circ - 132^\circ}{2} = 24^\circ$ (1); $\angle ACB = \frac{1}{2}\angle AOB = 66^\circ$ (1); $\angle ADB = 180^\circ - 66^\circ = 114^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Half-chord $= 15$ cm (1); Distance $d = \sqrt{17^2 - 15^2} = 8$ cm (1); Area $= \frac{1}{2} \times 30 \times 8$ (1); $= 120$ cm^2 (1); Base is the chord, height is the perpendicular (1).	1+1+1+1+1
	OR: $d_1 = \sqrt{25^2 - 7^2} = 24$ cm (1); $d_2 = \sqrt{25^2 - 24^2} = 7$ cm (1); The chord of length 48 cm is nearer to the centre (1); A longer chord lies nearer to the centre (2).	1+1+1+2
20	Half-chord $= 16$ cm (1); Distance $d = \sqrt{20^2 - 16^2} = 12$ cm (1); Area $= \frac{1}{2} \times 32 \times 12$ (1); $= 192$ cm^2 (1); Base is the chord, height is the perpendicular (1).	1+1+1+1+1
	OR: $d_1 = \sqrt{25^2 - 15^2} = 20$ cm (1); $d_2 = \sqrt{25^2 - 20^2} = 15$ cm (1); The chord of length 40 cm is nearer to the centre (1); A longer chord lies nearer to the centre (2).	1+1+1+2

Mistakes to avoid:

- Using the full chord instead of the half-chord in $r^2 = d^2 + (\text{half-chord})^2$.
- Forgetting that the perpendicular distance, not the slant distance, is used when finding the distance of a chord from the centre.
- Applying the angle theorems to an arc without checking the angle is at the *centre* (double) or on the *remaining* circle (half).
- For two parallel chords, adding the distances when they are on the same side or subtracting them when they are on opposite sides (the reverse is correct).
- Assuming that opposite angles of a cyclic quadrilateral are equal; they are supplementary (sum 180°).
- Using the angle in a semicircle theorem when the chord is not a diameter.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 68°	Central angle is double.
2	**(b)** 115°	Opposite angles of the cyclic quadrilateral are supplementary.
3	**(a)** 24	Equal chords are equidistant from the centre.
4	**(c)** 43°	Right angle at C.
5	**(b)** 16	Chord $= 2\sqrt{r^2 - d^2}$.
6	**(d)** 17	$r^2 = h^2 + d^2$.
7	**(a)** 28°	Isosceles triangle OAB.
8	**(d)** 105°	$\angle A + \angle C = 180^\circ$.
9	**(a)** 76°	Central angle is double.
10	**(d)** 105°	Opposite angles of the cyclic quadrilateral are supplementary.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle OAB = \frac{180-116}{2} = 32^\circ$ (1); $\angle ACB = 58^\circ$ (1).	1+1
12	Half-chord $= \sqrt{29^2 - 21^2} = 20$ cm (1); Chord $= 40$ cm (1).	1+1
13	$\angle C = 122^\circ$ (1); $\angle D = 89^\circ$ (1).	1+1
14	$\angle OAB = \frac{180-144}{2} = 18^\circ$ (1); $\angle ACB = 72^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Distances from the centre: 12 cm and 5 cm (1); Opposite sides: add (1); Distance $= 17$ cm (1).	1+1+1
16	$\angle A + \angle C = 180^\circ$; $6x = 168$ (1); $x = 28$ (1); $\angle A = 102^\circ$, $\angle C = 78^\circ$ (1).	1+1+1
17	$\angle OAB = \frac{180^\circ - 148^\circ}{2} = 16^\circ$ (1); $\angle ACB = \frac{1}{2}\angle AOB = 74^\circ$ (1); $\angle ADB = 180^\circ - 74^\circ = 106^\circ$ (1).	1+1+1
	OR: Distances from the centre: 15 cm and 8 cm (1); Opposite sides: add (1); Distance $= 23$ cm (1).	1+1+1
18	$\angle A + \angle C = 180^\circ$; $9x = 171$ (1); $x = 19$ (1); $\angle A = 88^\circ$, $\angle C = 92^\circ$ (1).	1+1+1
	OR: $\angle OAB = \frac{180^\circ - 156^\circ}{2} = 12^\circ$ (1); $\angle ACB = \frac{1}{2}\angle AOB = 78^\circ$ (1); $\angle ADB = 180^\circ - 78^\circ = 102^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Half-chord $= 24$ cm (1); Distance $d = \sqrt{25^2 - 24^2} = 7$ cm (1); Area $= \frac{1}{2} \times 48 \times 7$ (1); $= 168$ cm^2 (1); Base is the chord, height is the perpendicular (1).	1+1+1+1+1
	OR: $d_1 = \sqrt{10^2 - 6^2} = 8$ cm (1); $d_2 = \sqrt{10^2 - 8^2} = 6$ cm (1); The chord of length 16 cm is nearer to the centre (1); A longer chord lies nearer to the centre (2).	1+1+1+2
20	Half-chord $= 24$ cm (1); Distance $d = \sqrt{26^2 - 24^2} = 10$ cm (1); Area $= \frac{1}{2} \times 48 \times 10$ (1); $= 240$ cm^2 (1); Base is the chord, height is the perpendicular (1).	1+1+1+1+1
	OR: $d_1 = \sqrt{5^2 - 3^2} = 4$ cm (1); $d_2 = \sqrt{5^2 - 4^2} = 3$ cm (1); The chord of length 8 cm is nearer to the centre (1); A longer chord lies nearer to the centre (2).	1+1+1+2

Mistakes to avoid:

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- Forgetting that the perpendicular distance, not the slant distance, is used when finding the distance of a chord from the centre.
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